3. Method

3.1. Research design

The study follows a sequential observational design (Fig. 1). Theory informed vocabulary guides educational VTuber retrieval without restricting academic subjects (Sections 3.2 and 3.3). Descriptive metadata analysis characterizes the sample (Section 3.4). LDA identifies instructional topics in video metadata (Section 3.5), and automated sentiment and aspect extraction describes public comments (Section 3.6). The outputs are linked through video and comment identifiers. Section 3.7 then examines aspect associations within comments, their aggregation across videos, topics and formats, and the robustness of the resulting network. A neural graph model is not required for the primary exploratory research question.

[Figure to be inserted]

Fig. 1. Theory informed analytical sequence from broad retrieval to video metadata analysis, topic modeling, audience evaluation analysis, and relational network interpretation. The graph represents observable cooccurrence and contextual associations, not causal paths.

3.2. Theoretical basis for keyword selection

Three perspectives guided the selection of evaluation aspects (Table 1). The Community of Inquiry framework informed aspects of instruction, understanding, and social discussion (Garrison et al., 2000). The cognitive theory of multimedia learning informed pacing, visual presentation, and voice (Mayer, 2021). Parasocial interaction theory informed references to interpersonal connection with VTuber personas (Horton & Wohl, 1956; Dibble et al., 2016). These theories provide interpretive categories, not direct measures of psychological constructs.

Table 1. Theoretical perspectives and concepts guiding comment evaluation.

Theory	Key concepts	Relevance to VTuber learning	Comment evaluation aspect
Community of Inquiry (Garrison et al., 2000)	Teaching presence: design, instruction, facilitation (Anderson et al., 2001)	Organization of lessons, explanations, and responses to viewers	A1 organization; A2 explanation; A3 facilitation
	Cognitive presence: inquiry, exploration, integration, resolution (Garrison et al., 2001)	Questions and reported understanding or application	A4 understanding

Theory	Key concepts	Relevance to VTuber learning	Comment evaluation aspect
	Social presence: affective, interactive, and cohesive communication (Rourke et al., 1999)	Peer encouragement and discussion in comments	A5 peer interaction
Cognitive theory of multimedia learning (Mayer, 2021)	Segmenting; signaling and coherence; voice and personalization	Pacing, visual emphasis, and narration style	A6 pacing; A7 visual presentation; A8 voice and delivery
Parasocial interaction (Horton & Wohl, 1956; Dibble et al., 2016)	Perceived interaction; enduring perceived relationship	References to the avatar, affection, and companionship	A9 persona and avatar

Broad retrieval expressions combined VTuber identity and instructional activity, without specifying academic subjects, so that RQ1 can discover the content covered in the corpus (Table 2; Appendix A). Comment dictionaries contain illustrative evaluation expressions rather than exhaustive theory measures. Within A9, remarks about avatar appearance and expressions of companionship or attachment should be recorded separately when detectable, because visual preference is not equivalent to parasocial interaction.

Table 2. Representative keywords for video retrieval and comment evaluation.

Purpose	Key term or aspect	Example phrases
Video retrieval	VTuber identity	VTuber; virtual YouTuber
	Instructional activity	lesson; tutorial; explanation
Comment analysis	A1 organization; A2 explanation	well organized; clear explanation
	A3 facilitation; A4 understanding	answered my question; I understand
	A5 peer interaction; A6 pacing	study together; too fast
	A7 visual presentation; A8 voice and delivery	helpful diagram; clear voice
	A9 persona and avatar	love this avatar; keeps me company

3.3. Data collection

Candidate videos were retrieved through YouTube Data API v3 between [ ] and [ ] using [ ] queries (Appendix A). No subject terms restricted the searches. Because search results vary over time (Rieder et al., 2018, 2025), all queries and parameters were logged and searches were repeated on two dates. Two researchers verified the VTuber identity and educational purpose of candidate channels, and uploads from verified channels were screened to recover eligible lessons missed by search.

Videos were eligible when an identifiable avatar presenter provided explicit subject instruction, explanation, demonstration, or guided practice. Human hosted lessons, background only avatar appearances, fan edits and clips, advertisements, game streams without an instructional objective, and tutorials about VTuber software were excluded. Two researchers screened each candidate (κ = [ ]; Cohen, 1960) and resolved disagreements by discussion. Public metadata and all available top level comments and replies were collected for eligible videos. The final sample comprised [ ] channels, [ ] videos, and [ ] comments (Fig. 2).

Translation. Metadata and comments not written in English were machine translated into English with [translation service and version] before analysis, and the original language of each record was retained as a variable. Two bilingual researchers checked a random sample of [ ] translated comments per language for preservation of meaning, focusing on slang, irony, and VTuber specific expressions, and systematically mistranslated terms were corrected with a glossary.

The study analyzed publicly available content without interacting with creators or commenters and followed the Association of Internet Researchers guidelines (franzke et al., 2020). The protocol was reviewed by [ ] (approval number [ ]). Analytic files used coded identifiers with restricted access, and quoted comments are short and paraphrased where necessary to prevent identification through search.

[Figure to be inserted]

Fig. 2. Screening and data collection flow. Counts and exclusions will be populated from the query registry and screening records.

3.4. Descriptive metadata analysis

Table 3 lists the collected metadata and derived indicators. Textual metadata (titles, descriptions, tags, and chapter titles) formed the topic modeling corpus because the official API permits caption download only for users with permission to edit the video (Google for Developers, 2026a). Numerical metadata described the sample and served as node attributes in the network.

Because engagement accumulates with exposure, views, likes, and comments were divided by video age in days and log transformed. Dislike counts are not publicly available, and owners can hide like and subscriber counts, so hidden values were recorded as missing with an indicator.

Table 3. Metadata fields and derived indicators.

Level	Collected fields	Derived indicators and use
Channel	Channel identifier; creation date; description; declared country and language; subscriber count (if public); number of uploads	Channel age; log subscribers; share of lessons among uploads; network node attributes
Video (text)	Title; description; tags (if available); chapter titles	Cleaned document for LDA (RQ1)
Video (numeric and categorical)	Publication time; duration; category; caption availability; livestream archive or edited upload; original language; view, like, and comment counts; collection time	Video age; log views per day; likes per view; comments per view; duration; format (lecture, worked example, livestream lesson, short); network node attributes and attribute nodes
Comment	Comment, video, and parent identifiers; publication time; like count; channel owner authorship; original language; original and translated text	Input to sentiment analysis (RQ2); reply ratio and creator reply rate per video

3.5. Topic modeling of video metadata (RQ1)

Corpus and preprocessing. Each document consisted of the title, description, tags, and chapter titles of one video. Recurring channel specific boilerplate, such as social media links, sponsorship text, and membership announcements, was removed before modeling. English text, original or translated, was tokenized and lemmatized with a custom dictionary of subject terms and VTuber expressions. Stop words, URLs, numbers, emojis, and very rare and very frequent terms were removed, and frequent bigrams were added. All documents were modeled jointly, and topic stability was compared between originally English and translated documents (Appendix B).

Model. LDA assumes that each of K topics is a probability distribution over words and that each document is a mixture of topics with proportions θ (Blei et al., 2003). Models were estimated with online variational Bayes in gensim (Řehůřek & Sojka, 2010) over a grid of K values and several random seeds. The final K was chosen by topic coherence (Mimno et al., 2011; Röder et al., 2015), topic diversity (Dieng et al., 2020), stability across seeds, and inspection of top words and representative videos by two researchers (Chang et al., 2009; Sievert & Shirley, 2014). Perplexity was reported but not used for selection (Appendix B).

Interpretation. Two researchers labeled each topic from its most relevant words (Sievert & Shirley, 2014) and representative videos and resolved disagreements by discussion. Topics were grouped into kinds of learning, such as programming, mathematics, or language learning, and topics reflecting channel housekeeping were labeled as such. Each video was represented by its topic proportion vector θ, which was carried forward to the sentiment and network analyses.

3.6. Sentiment analysis of comments (RQ2)

Comment scope. All available public comments and replies on eligible videos were retained regardless of whether they mentioned instruction. They were treated as audience discourse associated with educational videos rather than as evidence that commenters were learning. Comments without a relevant aspect were retained in overall sentiment estimates.

Automated sentiment and aspect analysis. A documented pretrained English social media sentiment model (Barbieri et al., 2020) assigns overall polarity without training on study data. Transparent aspect dictionaries and contextual rules identify up to nine evaluation categories for each comment. Each aspect has an occurrence indicator and, where interpretable, its own polarity. Multiple aspects may be assigned to the same comment; comments with no detected aspect remain in the corpus. A9 additionally distinguishes avatar appearance from perceived companionship and enduring attachment where the language supports such classification. Negation, targets, and ambiguous phrases are addressed in versioned rules. The outputs should undergo a limited independent precision audit, even if the study does not build a manually annotated classifier. Without validation they remain provisional indicators of expressed evaluation.

Mapping responses to topics. Each comment is linked to the LDA topic proportions of its source video. Report weighted sentiment by topic, with valid denominators and channel clustered bootstrap intervals. Distinguish the percentage of comments mentioning an aspect from polarity conditional on mentioning it. Sensitivity checks compare original English comments with

translated comments and evaluate the influence of high volume channels. All comments remain eligible for overall sentiment analysis, but they are not presumed to report learning or satisfaction.

3.7. Theory informed relational network analysis (RQ3)

RQ3 examines three families of observable associations: instructional evaluation with reported understanding (A1 to A4), multimedia presentation with instructional evaluation (A6 to A8 with A1 to A4), and persona or interpersonal connection with instructional and peer evaluation (A9 with A1 to A5). The analysis asks whether these relationships are more prominent than expected from individual aspect prevalence and whether they vary by instructional topic and video format. Comment mentions do not measure teaching presence, cognitive processing, or parasocial bonds as latent constructs.

Graph construction. The heterogeneous network includes channel, video, topic, format, language, comment, and evaluation aspect nodes (Table 4; Fig. 3). An observed comment connects to its source video and to each aspect detected in its text, retaining its aspect polarity. Videos connect to their channels, metadata inferred topics, formats, and languages. This distinguishes two aspects appearing in the same comment from aspects mentioned separately under the same video. The comment by aspect incidence matrix is retained as an auditable representation. Aggregation to video and topic level preserves denominators and prevents highly commented videos from mechanically determining association strength.

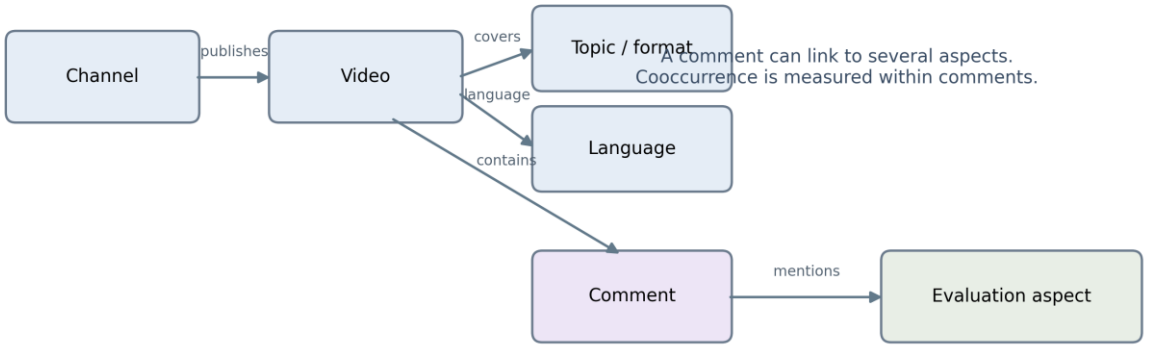


Fig. 3. Proposed heterogeneous network with comment nodes connecting individual audience responses to multiple evaluation aspects. Channel, video, topic, format, and language relations supply context. An aspect pair is linked at the comment level only when both are identified within one comment. Edges represent observations or computational assignments, not causal effects.

Table 4. Node and edge types in the heterogeneous network.

Element	Definition	Attributes or relation weight
Channel node	Verified educational VTuber channel	Channel metadata
Video node	Eligible educational video	Video metadata and LDA topic proportions
Topic node	Topic inferred from metadata	Topic label and probability
Format node	Lecture, worked example, livestream lesson, or short	Video format category
Language node	Original language	Video language category
Comment node	Individual public comment or reply	Video ID, polarity, aspect assignments; coded ID
Evaluation aspect node	One of A1 to A9	Theory informed indicator; A9 subtypes if detectable
publishes	Channel to video	Observed upload
contains	Video to comment	Observed comment relation
mentions	Comment to aspect	Mention indicator and conditional sentiment
covers / has format / uses language	Video to topic, format, language	Topic probability or categorical link

Network analysis and propositions. For each pair of aspects, the analysis reports comment level cooccurrence counts and proportions, an observed to expected association ratio or log odds ratio, and uncertainty estimated by resampling channels. Null models account for marginal aspect prevalence and, where feasible, the number of aspects per comment. A weighted aspect network summarizes within theory and between theory associations. Community detection is exploratory and is compared with null networks. Networks are then estimated across video formats, topic

families, and original languages; comparisons account for unequal comment volumes and channel clustering. The analysis distinguishes descriptive prominence from association beyond chance and applies correction where multiple pairwise tests are performed.

Optional graph neural network extension. The principal RQ3 analysis does not require training a neural model. A heterogeneous GNN may be added only if the data support a clearly independent prediction task, such as using earlier observations to predict future comment aspect distributions for previously unseen videos. Training and evaluation must separate source from target comments and prevent information flow from test outcomes into graph features. The model should be compared with metadata regression and strong non graph baselines under channel grouped or chronological validation. Improved accuracy would indicate predictive value of relational information, not validate a psychological mechanism. If an independent target or sufficient samples are unavailable, omit the GNN and report the transparent network analyses as the substantive RQ3 evidence.

Interpretation. Each salient relation should be traced to its comments, constituent videos, topic proportions, and contextual definitions. Instructional and persona aspects cooccurring in one comment provide stronger evidence of integrated discourse than their mere coexistence in a video comment section, but neither supports causal inference or individual learning gains. Stable patterns may motivate theoretical propositions about avatar mediated learning, whereas weak or absent associations are equally informative. The distinctive role of VTubers cannot be established without a comparison with conventional presenters.

4. Results

ILLUSTRATIVE TEMPLATE. All values in Section 4 are simulated solely to show table and prose conventions. Replace them with reproducible empirical outputs, uncertainty intervals, and validated interpretation.

4.1. Overview of the data

Hypothetical example: 48 channels, 640 videos, and 18,000 comments. Report the number of comments with a supported language and usable text, videos with comments, comments mentioning each aspect, and any inaccessible records.

Table 5. Sample overview (synthetic template).

Indicator	Illustrative value
Channels	48
Eligible videos	640
Retrieved comments	18,000
Comments excluded for learning relevance	0 (all retained)
Comments with supported language and usable text	15,500

4.2. Instructional topics in educational VTuber videos (RQ1)

Hypothetical example: LDA identifies four topic families. Actual topic counts and labels must be checked against representative videos and for stability between originally English and translated documents.

Table 6. Learning topic families (synthetic template).

Topic family	Illustrative share	Illustrative keywords
Programming and computing	34%	Python; coding; algorithm
Mathematics and science	26%	calculus; statistics; physics
Language learning	24%	English; Japanese; vocabulary
Arts and humanities	16%	drawing; music; history

4.3. Audience evaluations by learning topic (RQ2)

4.3.1. Overall sentiment by learning topic

Hypothetical example: 55% of comments are positive, 17% negative, and 28% neutral. These proportions describe the tone of public comments, not learning achievement or satisfaction. Recalculate Table 7 from actual predictions.

Table 7. Overall sentiment across topic families (synthetic template).

Topic family	Positive	Negative	Neutral
Programming and computing	52%	20%	28%
Mathematics and science	50%	19%	31%
Language learning	63%	11%	26%
Arts and humanities	56%	17%	27%

4.3.2. Aspect level sentiment by learning topic

Hypothetical example: comments on language lessons may be more positive about voice and delivery, whereas comments on programming may more often concern explanation clarity and unresolved

understanding. Report aspect prevalence, sentiment conditional on mention, and sensitivity results for every aspect.

Table 8. Aspect patterns (synthetic template).

Aspect	Illustrative pattern	Required statistic
Explanation	More requests for clear worked steps in computing	Mention count; polarity; CI
Understanding	More unresolved questions in mathematics	Resolved and unresolved counts
Peer interaction	Peer encouragement varies by lesson format	Prevalence; sentiment; CI
Voice and delivery	More positive voice comments in language lessons	Prevalence; sentiment; CI
Persona and avatar	Praise not necessarily linked to comprehension	Prevalence; sentiment; CI

4.4. Connections among instructional, multimedia, and persona evaluations (RQ3)

4.4.1. Comment level association patterns

ILLUSTRATIVE REPORTING TEMPLATE. Report the total number of comments with two or more detected aspects, pairwise cooccurrence counts, expected frequencies, normalized association ratios, channel bootstrap confidence intervals, and adjusted comparisons against the null model. Separate pairs drawn from one theoretical domain from cross domain pairs. The hypothetical values in Table 9 are not empirical findings.

Table 9. Illustrative comment level associations normalized by aspect prevalence (all values synthetic).

Comment aspect pair	Observed to expected ratio (synthetic)	Theoretical lens
Explanation and understanding	1.48	Within CoI indicators
Voice and explanation	1.21	CTML with CoI indicators
Persona and peer interaction	1.36	PSI with CoI indicators
Persona and understanding	1.09	PSI with expressed understanding
Pacing and visual presentation	1.27	Within CTML indicators

4.4.2. Contextual variation and robustness

ILLUSTRATIVE REPORTING TEMPLATE. Compare aspect pair associations across topic families, video formats and original languages. Report uncertainty and the effect of high volume channels, alternative dictionary thresholds, and translated text. Table 10 contains synthetic values for format only.

Table 10. Illustrative sensitivity of the persona and peer interaction association (all values synthetic).

Specification	Illustrative ratio	Interpretation
Full comment corpus	1.36	Persona and peer interaction pair
Exclude largest channel	1.29	Channel concentration sensitivity
Strict aspect dictionary	1.24	Detection rule sensitivity

Specification	Illustrative ratio	Interpretation
Originally English comments	1.41	Language subset
Translated comments	1.18	Translation sensitivity

4.4.3. Relational communities and theoretical configurations

ILLUSTRATIVE REPORTING TEMPLATE. Report whether robust communities separate instructional, multimedia and persona aspects or connect them. Compare observed configurations with degree or margin preserving null models. A persona cluster that remains distinct from instructional evaluations would challenge claims of integrated educational persona experiences; a stable cross domain community would motivate, but not prove, a model of interconnected evaluations.

4.4.4. Optional independent prediction test

Do not report trained GNN performance unless an independent prediction task is implemented. If earlier comment signals predict a later and separate audience response window, Table 11 can summarize model comparisons with channel grouped evaluation and confidence intervals. Otherwise replace the table with a clear statement that RQ3 was addressed by association and network evidence without neural training.

Table 11. Conditions for an optional supplementary graph prediction analysis (no empirical performance reported).

Supplementary test	Independent target	Comparison	Reporting condition
Earlier to later comments	Later aspect mention profile	Metadata only versus graph	Only if timestamps and sufficient sample
Unseen video prediction	Held out later videos	Regression versus trained GNN	Strict feature and channel isolation
No defensible target	Not applicable	No GNN necessary	Primary network analysis is sufficient
Model interpretation	No causal target	Uncertainty and ablations	Exploratory evidence only

Appendix A. Video search protocol and keywords

Search terms combine a presenter identity expression, such as VTuber or virtual YouTuber, with a general instructional activity expression, such as lesson, tutorial, or explained. Equivalent expressions in [other languages] are listed in the supplementary query registry. Academic disciplines are excluded from queries because identifying them is the objective of RQ1. Query logs and exclusion decisions are retained, and optional exclusion terms are evaluated in pilot searches to avoid removing relevant lessons.

A.1. Keyword families

Table A1. Search keyword families.

Family	Examples
Presenter identity	VTuber; virtual YouTuber; virtual teacher
Instructional activity	lesson; tutorial; teaching; explained; lecture
Pilot exclusions only	Live2D; rigging; avatar creation; fan clip

Exclusion terms are tested in the pilot rather than applied to every query, because some legitimate lessons mention software or other excluded expressions.

A.2. Illustrative queries

Table A2 lists illustrative queries; the registered search list, dates, parameters, and results will be reported after collection. The OR operator (|) is URL encoded as %7C when sent to the API (Google for Developers, 2026b).

Table A2. Illustrative broad search queries without subject restrictions.

No.	Illustrative query	Purpose
Q1	VTuber (lesson|tutorial|teaching)	Broad video discovery
Q2	"virtual YouTuber" (lesson|explained|lecture)	Alternative identity wording
Q3	"VTuber teacher"|"educational VTuber"	Channel discovery
Q4 to Qn	Equivalent queries in [other languages]	Retrieval of videos in other languages for translation

A.3. API parameters

Video searches used the search.list endpoint with type = video, maxResults = 50, and pagination through nextPageToken. English queries used relevanceLanguage = en with regionCode = US and GB, and queries in other languages used the corresponding relevanceLanguage and regionCode values. Each query was run with order = relevance and order = date, and the search period was divided into yearly windows with publishedAfter and publishedBefore so that older videos were not crowded out by recent ones. Channel discovery used Q3 and equivalent identity expressions with type = channel. Because search.list is limited to 100 calls per day (Google for Developers, 2026b), queries were scheduled over several days, beginning with channel discovery. Uploads of verified channels were retrieved through the uploads playlist (channels.list and playlistItems.list), metadata through videos.list, and comments through commentThreads.list and comments.list.

A.4. Channel discovery and stopping rule

Channels were also discovered by snowball sampling: featured channels and collaboration partners of each verified channel were inspected and added when they met the verification criterion. Query variants were added during the pilot until a new variant contributed fewer than 5% new eligible videos, at which point the keyword set was frozen.

A.5. Records kept

For every query run, the registry stored the query identifier, q string, parameters, date and time, page tokens, number of returned items, and video or channel identifiers. For every candidate, the screening log stored the identifier, title, channel, decision, exclusion reason, and coder. Full registry and screening records: [ ]